

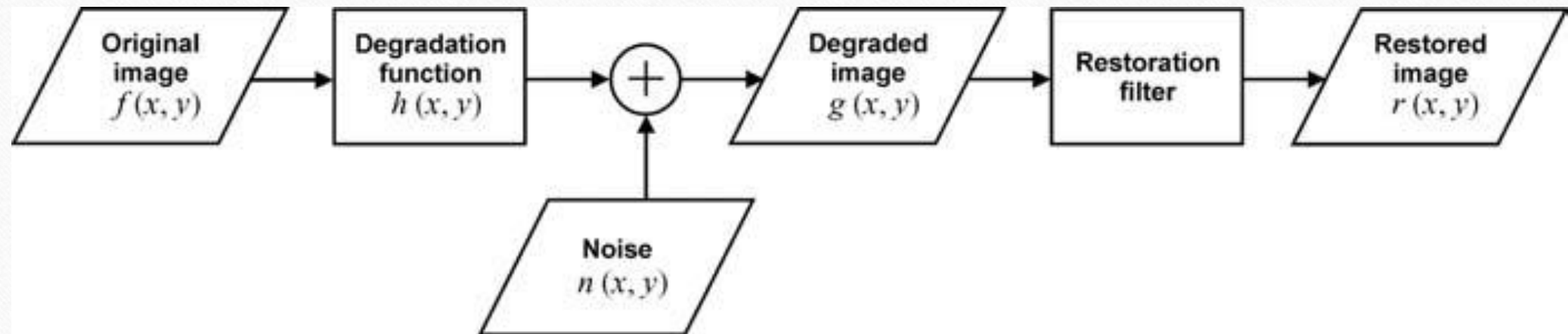
CHAPTER 12

IMAGE RESTORATION

WHAT WILL WE LEARN?

- What is noise (in the context of image processing) and how can it be modeled?
- What are the main types of noise that may affect an image?
- What is blurring (in the context of image processing) and how can it be modeled?
- Which noise removal techniques are typically used in image processing?
- Which deblurring techniques are typically used in image processing?

IMAGE DEGRADATION AND RESTORATION PROBLEM



$$g(x, y) = f(x, y) * h(x, y) + n(x, y)$$

where $*$ denotes convolution

$$G(u, v) = F(u, v)H(u, v) + N(u, v)$$

Restoration filter

1. Collect knowledge about the degradation process
2. Use that knowledge to develop a degradation model.
3. Develop the inverse degradation process and model it as a filter.

NOISE AND NOISE MODELS

- Gaussian Noise
- Impulse (Salt and Pepper) Noise
- Uniform Noise
- Rayleigh Noise
- Gamma (Erlang) Noise
- Exponential Noise

Noise Estimation

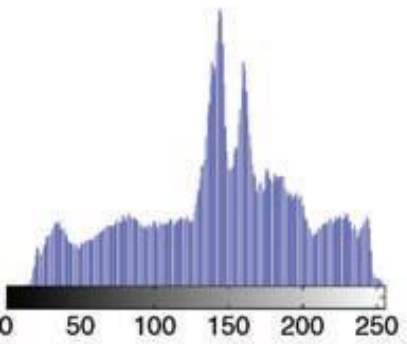
- It is often necessary to estimate the type of noise (and the main parameters of its PDF) before designing a solution for noise reduction.
 - If primary source of noise is sensor: generate test images consisting of large patches of homogeneous gray-level values and observe the resulting histograms
 - Crop a relatively large homogeneous region from an image and inspect its histogram



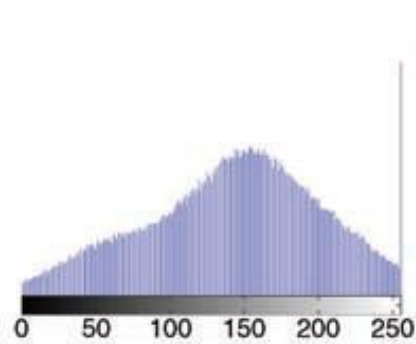
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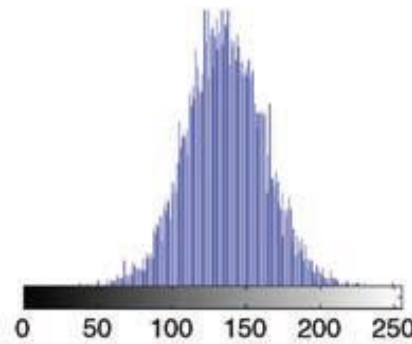
(b)



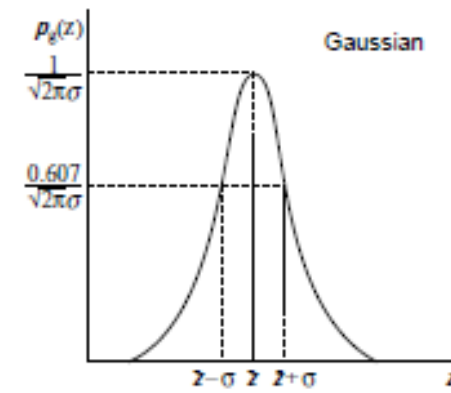
(c)



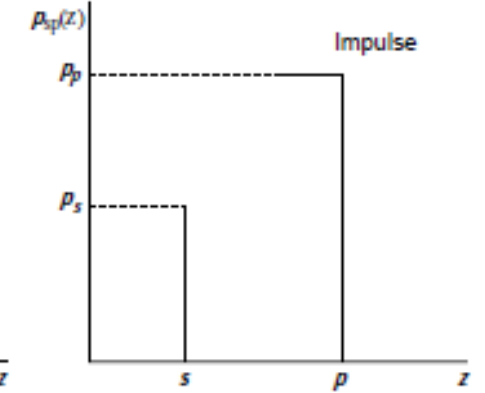
(d)



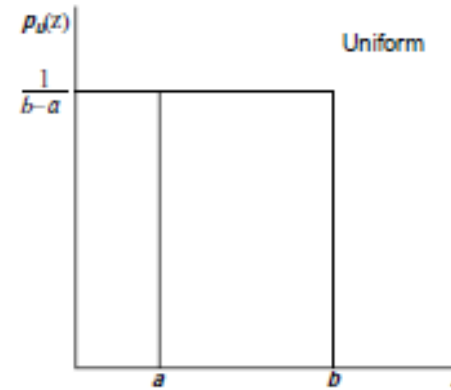
(e)



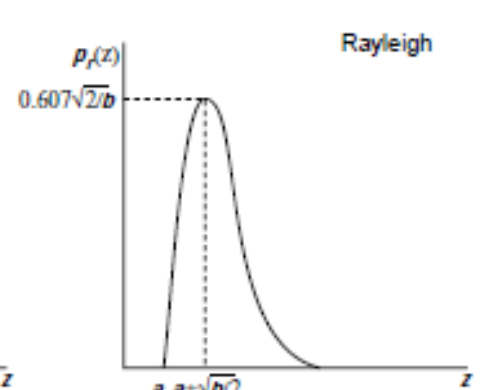
(a)



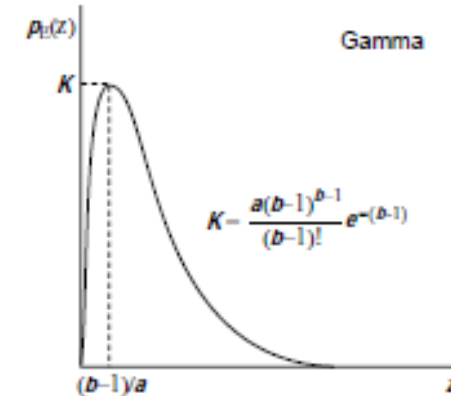
(b)



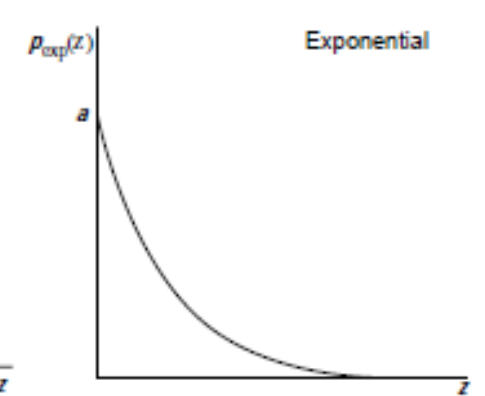
(c)



(d)

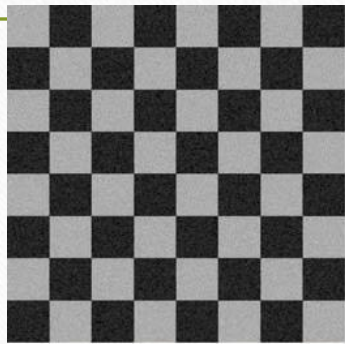


(e)

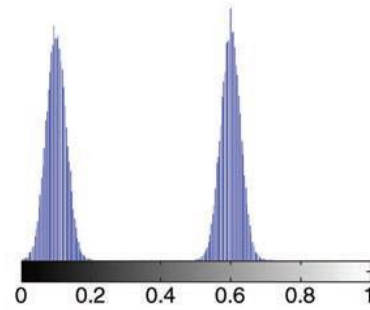


(f)

(a) original image; (b) noisy image (c) histogram of the original image;
 (d) histogram of the noisy image; (e) histogram of selected patch
 showing clearly that the noise is of Gaussian type in this case.

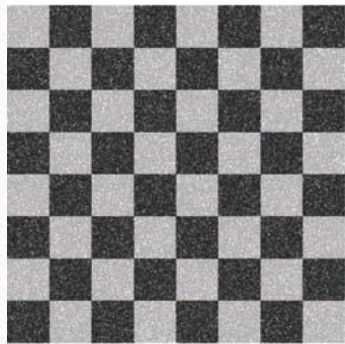


(a)

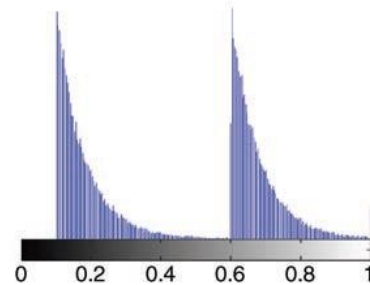


(b)

(a and b)
Gaussian;

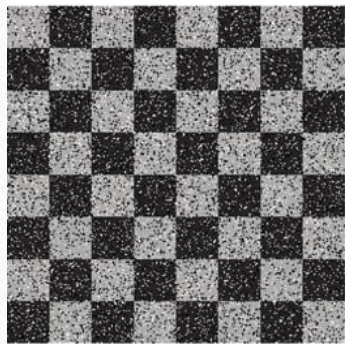


(c)

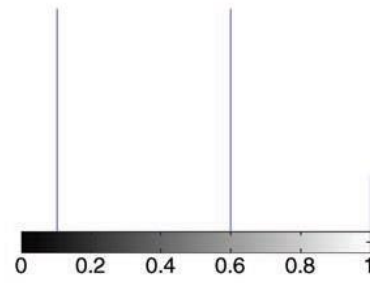


(d)

(c and d)
exponential;

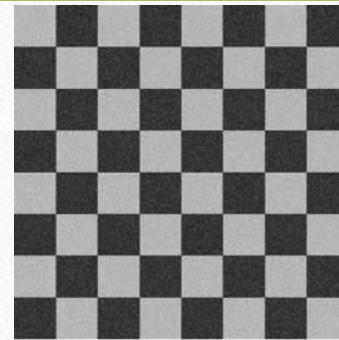


(e)

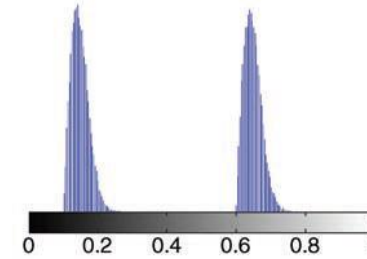


(f)

(e and f)
salt
and pepper.

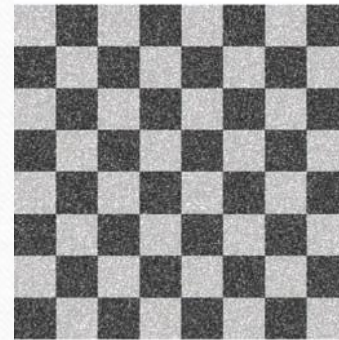


(a)

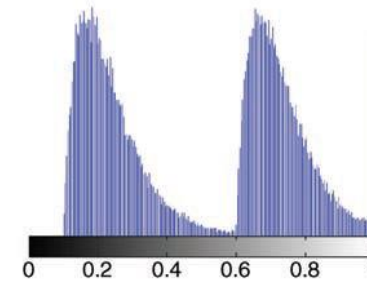


(b)

((a and b)
Rayleigh;

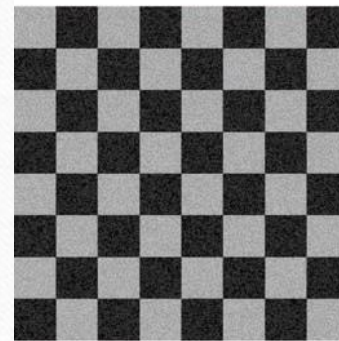


(c)

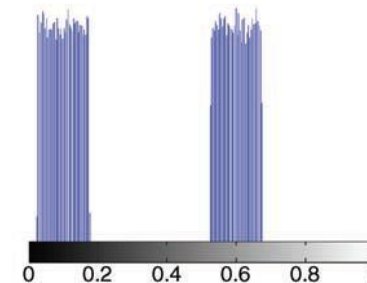


(d)

(c and d)
Gamma;



(e)



(f)

(e and f)
uniform.

Image Restoration Techniques

1. NOISE REDUCTION USING SPATIAL-DOMAIN TECHNIQUES
2. NOISE REDUCTION USING FREQUENCY-DOMAIN TECHNIQUES
3. IMAGE DEBLURRING TECHNIQUES

NOISE REDUCTION USING SPATIAL-DOMAIN TECHNIQUES

- **Mean Filters**

- **Arithmetic Mean Filter:** operates on an $m \times n$ sliding window by calculating the average of all pixel values within the window and replacing the center pixel value in the destination image with the result
- Geometric Mean Filter
- Harmonic Mean Filter
- Contraharmonic Mean Filter

- **The Median Filter:** selecting the middle pixel value from the ordered set of values within the $m \times n$ neighborhood (W) around the reference pixel.

In Matlab

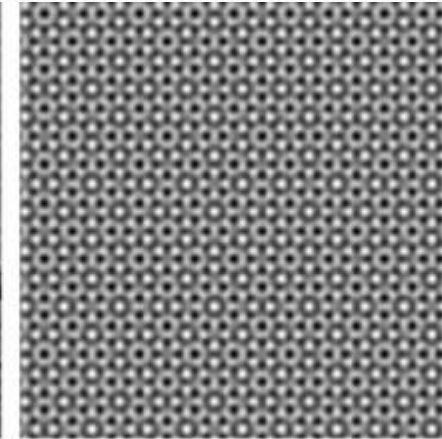
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NOISE REDUCTION USING FREQUENCY DOMAIN TECHNIQUES

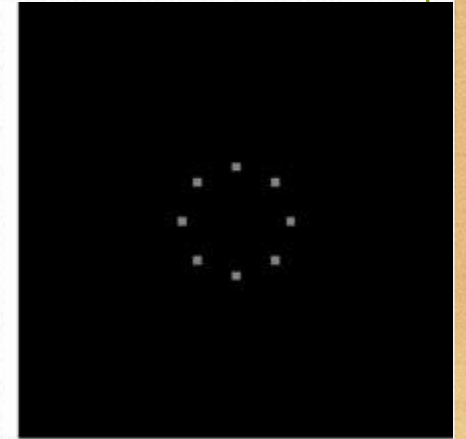
- Periodic Noise: arises as a result of electrical or electromechanical interference during the image acquisition process.



(a)



(b)



(c)

Bandreject Filter

- attenuates frequency components within a certain range (the stopband of the filter), while leaving all other frequency components untouched

Bandpass Filter

- Allows certain frequencies (within its passband) to be preserved while attenuating all others

IMAGE DEBLURRING TECHNIQUES

$$G(u, v) = F(u, v)H(u, v) + 0$$

$$F(u, v) = \frac{G(u, v)}{H(u, v)} = G(u, v) \frac{1}{H(u, v)}$$

Wiener Filtering

- Is an image restoration solution that can be applied to images that have been subject to a degradation function and also contain noise

$$R(u, v) = \left[\frac{1}{H(u, v)} \frac{|H(u, v)|^2}{|H(u, v)|^2 + K} \right] G(u, v)$$

K is a constant used to approximate the amount of noise

In MATLAB

- **deconvreg**: deblur image using regularized filter
- **deconvlucy**: deblur image using Lucy–Richardson method
- **deconvblind**: deblur image using blind deconvolution

Hands-on

- Tutorial 12.1
- Problems